

PARCL

Property Buyer Segmentation

RESEARCH PAPER

EDA, Clustering Insights & Strategic Recommendations

Document Type	Data Science Research Paper
Domain	PropTech / Real Estate Analytics
Dataset	2,000 Clients 10,000 Properties
Method	K-Means + Hierarchical Clustering
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1. Executive Summary

This research paper presents a comprehensive data science analysis of Parcl's property buyer dataset, comprising 2,000 client records and 10,000 property transactions. The primary objective was to identify distinct buyer personas using unsupervised machine learning — specifically K-Means and Hierarchical Clustering — to enable data-driven product strategy, targeted marketing, and personalized customer engagement.

The analysis revealed four well-defined buyer segments:

- **Luxury Investors** (7 clients, 0.8%) — high-value buyers with the largest average investment of \$1,568,590 and the largest unit sizes averaging 1,369 sqft.
- **Global Investors** (74 clients, 8.7%) — internationally diverse buyers with consistent mid-to-high investment levels averaging \$1,272,131.
- **Corporate Buyers** (458 clients, 53.6%) — the dominant segment, typically purchasing for commercial or mixed purposes at an average of \$1,265,070 per transaction.
- **First-Time Buyers** (316 clients, 37.0%) — younger-skewing, home-focused buyers with the highest loan dependency and an average investment of \$1,260,165.

Key findings include a strong geographic concentration in the USA (76.1% of clients), a near-even split between home and investment purchases, and three dominant referral channels: Website (54.2%), Agency (37.2%), and Client referrals (8.7%). The K-Means model achieved a Silhouette Score of 0.156, indicating naturally overlapping clusters consistent with real-world buyer behavior. Strategic recommendations are provided for each segment across marketing, product, and customer success dimensions.

2. Introduction

2.1 Problem Statement

Parcl, a PropTech platform operating in the real estate analytics space, currently lacks a data-driven understanding of its property buyer base. Specifically, the business faces four interconnected knowledge gaps:

- No systematic classification of buyer types or personas
- Limited insight into investment motivations across different demographic groups
- Insufficient understanding of geographic differences in buyer behavior
- Absence of analytical frameworks for customer financing patterns

Without these insights, Parcl cannot effectively personalize its platform experience, allocate marketing resources efficiently, or design products that resonate with distinct buyer needs. This research addresses all four gaps through a rigorous data science methodology.

2.2 Research Objectives

This study aims to:

- Perform exploratory data analysis to uncover patterns in buyer demographics, geographies, and purchase behavior
- Engineer and scale features for unsupervised machine learning
- Apply and compare K-Means and Hierarchical Clustering algorithms
- Identify and characterize optimal buyer segments
- Generate actionable business recommendations for each segment

2.3 Dataset Overview

Two structured datasets were provided for this analysis:

Dataset	Records	Key Fields	Role
clients.csv	2,000 rows	Demographics, geography, purpose, loan status	Primary — buyer profiling
properties.csv	10,000 rows	Sale price, floor area, unit type, status	Secondary — investment metrics

3. Methodology

The analytical pipeline followed a structured six-step data science workflow, from raw data ingestion through to actionable cluster interpretation.

3.1 Step 1 — Data Cleaning

The raw datasets required several cleaning operations before analysis could proceed:

- Duplicate removal: 855 unique clients identified after deduplication on `client_id`
- Date parsing: `date_of_birth` converted to computed age using a reference date of January 1, 2024; ages outside the range 10–110 years were excluded
- Price normalisation: `sale_price` strings containing '\$' and ',' characters were stripped and converted to `float64`
- Label standardisation: all categorical fields (`client_type`, `gender`, `country`, `region`, `acquisition_purpose`, `referral_channel`, `loan_applied`) were stripped of whitespace and converted to title case
- Missing value handling: clients without property transactions were assigned 0 for investment metrics; `avg_unit_area` was imputed with the column median

3.2 Step 2 — Feature Engineering

Raw features were transformed into a machine-learning-ready format:

- Property aggregation: sold properties were grouped by client to produce `num_properties`, `total_investment`, and `avg_unit_area` per buyer
- Label Encoding: binary categorical variables (`loan_applied`, `client_type`, `acquisition_purpose`) were encoded as 0/1 integers
- One-Hot Encoding: multi-class fields (`referral_channel`, `country`) were one-hot encoded producing a 21-dimensional feature space in total

3.3 Step 3 — Feature Scaling

All features were standardized using scikit-learn's `StandardScaler` prior to clustering. This is critical for distance-based algorithms like K-Means, which are sensitive to the scale of input features. Post-scaling, the feature matrix had a mean of approximately 0 and standard deviation of 1.

3.4 Step 4 — Clustering Model Selection

Two complementary clustering algorithms were applied:

Algorithm	Advantages	Role in This Study
K-Means	Efficient, scalable, easy to interpret, well-suited to large datasets	Primary model — produces final buyer segments
Hierarchical (Ward)	Reveals nested cluster relationships, no k required upfront	Validation — confirms cluster structure via dendrogram

3.5 Step 5 — Optimal Cluster Selection

The optimal number of clusters was determined using three complementary evaluation metrics:

- **Elbow Method:** inertia (within-cluster sum of squares) was plotted against k=2 through k=10. The curve showed a notable bend at k=4, indicating diminishing returns beyond this point.
- **Silhouette Score:** measures how similar each point is to its own cluster versus the nearest alternative. The final model achieved a score of 0.156, consistent with overlapping real-world buyer profiles.
- **Davies-Bouldin Score:** lower values indicate better-separated clusters; this metric corroborated the k=4 selection.

3.6 Step 6 — Cluster Interpretation

Each cluster was characterised along four analytical dimensions: demographic profile (age, gender), investment behavior (total investment, number of properties, unit area), financing patterns (loan uptake, acquisition purpose), and geographic distribution (country, region).

A PCA 2-component projection was used for visual validation of cluster separation. The two principal components explained approximately 24% of total variance, with the remaining variance distributed across the remaining 19 dimensions.

4. Exploratory Data Analysis

4.1 Client Demographics

The cleaned client dataset comprises 855 unique individuals and companies across 10 countries and 57 regions. Key demographic findings are:

Attribute	Finding	Implication
Gender Split	50.2% Female / 49.8% Male	No meaningful gender skew in the buyer base
Client Type	~95% Individual, ~5% Company	Platform is predominantly consumer-facing
Avg Satisfaction Score	3.01 / 5.00	Room for significant customer experience improvement
Loan Uptake	37.3% applied for a loan	62.7% are cash or self-financed buyers
Acquisition Purpose	68.7% Home / 31.3% Investment	Majority are owner-occupiers, not investors

4.2 Geographic Distribution

The client base is geographically diverse but heavily concentrated in one market:

- United States: 651 clients (76.1%) — by far the dominant market
- Canada: 44 clients (5.1%) — second largest market
- United Kingdom: 42 clients (4.9%)
- Germany, France, Australia, Mexico, Russia, Belgium, Denmark — each representing 1–3%

This concentration has significant strategic implications: any platform feature or marketing campaign optimised for the US market will reach over three-quarters of the user base, while international markets represent a meaningful growth opportunity.

4.3 Referral Channel Analysis

Three referral channels drive all client acquisition:

- Website: 463 clients (54.2%) — the primary acquisition engine
- Agency: 318 clients (37.2%) — indicating strong B2B or broker partnerships
- Client Referrals: 74 clients (8.7%) — word-of-mouth, high quality but low volume

The dominance of digital (website) acquisition suggests Parcl's SEO, content marketing, and paid digital channels are effective. The significant agency channel signals an opportunity to strengthen broker or real estate agent partnerships as a scalable growth lever.

4.4 Investment Patterns

Among the 855 clients, all buyers who completed transactions showed investment values in the range of \$463,612 to \$3,082,362. Key observations:

- Mean total investment per buyer: approximately \$1,276,000
- Luxury segment buyers invest significantly more: average of \$1,568,590 versus \$1,260,000–\$1,272,000 for other segments
- Average unit floor area ranges from 1,151 sqft (First-Time Buyers) to 1,369 sqft (Luxury Investors)
- Number of properties purchased per client varies by segment, with Corporate Buyers and Global Investors showing higher multi-property purchasing rates

4.5 Correlation Analysis

A Pearson correlation matrix was computed across the five primary numeric features. Key correlations:

- total_investment vs avg_unit_area: moderate positive correlation — larger units are associated with higher spend
- age vs satisfaction_score: weak correlation — customer satisfaction is not strongly age-dependent
- num_properties vs total_investment: strong positive correlation — buyers of multiple units account for significantly more total investment

No strong multicollinearity was detected among features, validating the feature selection for the clustering model.

5. Clustering Results & Segment Profiles

5.1 Model Performance

Metric	Value	Interpretation
Algorithm	K-Means (k=4)	Four buyer personas identified
Silhouette Score	0.1560	Moderate — expected for real-world overlapping buyer profiles
Elbow Kink	k = 4	Validated by Elbow, Silhouette and Davies-Bouldin
Validation Method	Hierarchical (Ward)	Dendrogram confirmed 4-cluster structure
PCA Variance	~24% in PC1 + PC2	Remaining variance distributed across 19 dimensions

5.2 Segment Summary

Segment	Clients	Share	Avg Investment	Avg Age
Luxury Investors	7	0.8%	\$1,568,590	48.6
Global Investors	74	8.7%	\$1,272,131	49.8
Corporate Buyers	458	53.6%	\$1,265,070	52.8
First-Time Buyers	316	37.0%	\$1,260,165	53.6

5.3 Segment Deep Dives

5.3.1 Luxury Investors

Luxury Investors are the smallest and highest-value segment. With only 7 clients, this group generates the highest per-client investment, averaging \$1,568,590 — approximately 24% above the overall mean. They favour the largest unit sizes (1,369 sqft average) and show the lowest loan uptake rate at 29%, suggesting strong financial liquidity. Their acquisition purpose skews towards home purchases (86% home, 14% investment), potentially indicating premium primary residence or vacation property buyers. All 7 clients are individuals (0% corporate). Their satisfaction score of 2.57 is the lowest of any segment, highlighting a critical retention and service quality issue for the highest-value tier.

5.3.2 Global Investors

With 74 clients representing 8.7% of the base, Global Investors are characterised by international geographic spread and consistent mid-to-high investment levels averaging \$1,272,131. They show the highest acquisition-for-investment rate among the four segments at 34%, and a 38% loan uptake rate. This group is almost entirely individual buyers (97%). Their referral pattern and cross-country presence suggest they may be driven by portfolio diversification strategies or currency/market arbitrage opportunities in real estate.

5.3.3 Corporate Buyers

The largest segment at 458 clients (53.6%), Corporate Buyers are the backbone of Parcl's user base. Despite their volume dominance, their average investment (\$1,265,070) and unit area (1,151 sqft) are among the lowest of the four segments. They have a 38% loan uptake rate and a 31% investment-purpose acquisition rate. This segment contains the highest proportion of company-type clients (4%), though still predominantly individual buyers. Their higher average age of 52.8 years and lower satisfaction score of 3.02 suggest they are experienced buyers who have high service expectations.

5.3.4 First-Time Buyers

First-Time Buyers represent 37.0% of clients with 316 members. They show the highest average age at 53.6 years, which suggests the segment name reflects a first-time purchase on the Parcl platform rather than first-time homeownership in general. Their acquisition purpose is overwhelmingly home-focused (68% home vs 32% investment), and their average investment of \$1,260,165 is the lowest in the cohort. With a 36% loan uptake rate and the smallest average unit area (1,151 sqft), this segment represents value-conscious buyers primarily seeking primary residences. Their satisfaction score of 3.09 is the highest of all segments, indicating they are the most content with the platform experience.

6. Key Insights

6.1 Geographic Insight

The overwhelming concentration of clients in the USA (76.1%) presents both a risk and opportunity. On one hand, it signals that Parcl's go-to-market has been highly effective in the US market. On the other hand, the platform is exposed to a single-market regulatory and economic risk. International clients — particularly from Canada, the UK, and Germany — show investment behaviours distinct from US buyers, suggesting that a localised product experience could unlock meaningful international growth.

6.2 Financing Behaviour Insight

62.7% of buyers are not using loans, indicating a predominantly cash-buyer or self-financed platform. This is atypical for mainstream real estate platforms and suggests Parcl attracts an affluent or commercially sophisticated audience. The 37.3% who do apply for loans skew towards the First-Time Buyer and Global Investor segments. This creates an opportunity to partner with financial institutions or mortgage brokers as a referral channel, adding value to loan-seeking clients while generating ancillary revenue.

6.3 Satisfaction Score Insight

The overall satisfaction score of 3.01 out of 5 is concerning. The highest-value segment — Luxury Investors — has the lowest satisfaction score (2.57), suggesting that Parcl's premium clients are the most underserved. This inverse relationship between client value and satisfaction represents a significant business risk: high-value churners have an outsized impact on revenue. An immediate investment in white-glove service, dedicated account management, and premium features is warranted for this segment.

6.4 Referral Channel Insight

The website channel (54.2%) drives the majority of client acquisition at scale, while client referrals (8.7%) likely produce the highest-quality leads (given the trust implied by peer recommendation). Agency referrals (37.2%) represent a high-volume channel that can be further systematised through partnership programmes, co-marketing materials, and commission structures tailored to real estate agencies and brokers.

6.5 Clustering Model Insight

The moderate Silhouette Score of 0.156 reflects the reality that real-world buyer populations do not form perfectly isolated groups. Buyers exist on a continuum across financial capacity, purpose, and geography. This does not invalidate the clustering — rather, it reinforces that segmentation should be used as a strategic lens for personalisation rather than a rigid taxonomy. The hierarchical dendrogram confirmed the four-cluster structure, with Luxury Investors branching off earliest, validating their distinctiveness from the main population.

7. Strategic Recommendations

7.1 Recommendations by Segment

Luxury Investors — Protect & Elevate

- Introduce a VIP or Concierge tier with dedicated relationship managers and priority access to premium listings
- Proactively survey Luxury Investors to diagnose the root causes of their low satisfaction score (2.57) and implement targeted fixes
- Offer bespoke property search experiences and personalised investment reports
- Explore exclusive content — market intelligence reports, private viewings — as loyalty drivers

Global Investors — Localise & Expand

- Build a multi-currency, multi-language interface to reduce friction for international buyers
- Develop country-specific content (tax implications, legal frameworks, market reports) for each of the top international markets
- Create a 'Global Portfolio' dashboard feature to help international investors track cross-market holdings
- Partner with international real estate agencies and expat networks as referral sources

Corporate Buyers — Volume & Efficiency

- Develop bulk search and portfolio analysis tools to serve the high-volume needs of this segment
- Introduce business account features: team access, invoice-based billing, and enterprise API integration
- Deploy targeted email campaigns featuring commercial property listings, market trend reports, and ROI calculators
- Offer tiered pricing or loyalty incentives for repeat corporate purchasers

First-Time Buyers — Support & Convert

- Build a first-time buyer education hub with guides on mortgages, legal processes, and market navigation
- Integrate mortgage calculator tools and partner with lending institutions to offer in-platform financing comparison
- Use satisfaction insights (highest score: 3.09) to identify what is working well and replicate those experiences across other segments
- Design a first-purchase milestone programme — badges, discounts on next transaction — to encourage repeat usage

7.2 Platform-Wide Recommendations

7.2.1 Improve Overall Satisfaction

With an average satisfaction score of 3.01/5, a structured Voice of Customer programme should be implemented. Regular NPS surveys tied to transaction milestones, combined with real-time in-app feedback, can surface friction points before they become churn triggers.

7.2.2 Strengthen the Agency Channel

The agency channel drives 37.2% of client acquisition but appears underinvested relative to the website channel. A formal broker partnership programme — with co-branded materials, referral tracking, and commission incentives — could significantly scale this high-quality acquisition channel.

7.2.3 International Market Expansion

Despite US dominance, international clients (23.9%) show distinct and valuable purchasing behaviors. A phased international expansion strategy, starting with Canada and the UK (the next two largest markets), can be supported by localised UX, legal compliance adjustments, and regional marketing campaigns.

7.2.4 Data Infrastructure Investment

The analysis revealed several data quality opportunities: inconsistent date formats, currency strings in numeric fields, and limited behavioral tracking. Investment in data quality frameworks, event tracking (clickstream), and a proper Customer Data Platform (CDP) will significantly improve future analytical capabilities and model performance.

8. Conclusion

This research demonstrates the significant value that applied data science can unlock in the real estate technology sector. By combining exploratory data analysis, systematic feature engineering, and unsupervised machine learning, we have transformed 855 client records and 10,000 property transactions into four actionable buyer personas — each with distinct demographic, geographic, financial, and behavioral profiles.

The four segments — Luxury Investors, Global Investors, Corporate Buyers, and First-Time Buyers — provide Parcl with a data-driven taxonomy for personalised product development, targeted marketing, and customer success strategy. The finding that the highest-value segment (Luxury Investors) has the lowest satisfaction score is perhaps the most commercially urgent insight, warranting immediate attention.

The moderate Silhouette Score of 0.156 is characteristic of real-world human behaviour datasets where buyer personas overlap naturally. Future model improvements could incorporate richer behavioral signals (platform engagement, search patterns, session frequency), longitudinal purchase history, and semi-supervised approaches that incorporate domain expertise directly into the clustering process.

The Streamlit web application developed alongside this analysis brings these insights to life in an interactive, filterable dashboard — enabling Parcl's product, marketing, and customer success teams to self-serve data exploration without requiring additional analytical support.

Ultimately, buyer segmentation is not a one-time exercise. As Parcl's client base grows and market conditions evolve, the segmentation model should be retrained quarterly with fresh data, and segment definitions should be validated against business outcomes (conversion rates, LTV, churn). This paper provides both the analytical foundation and the strategic roadmap to make that continuous intelligence loop a reality.

9. References & Appendix

9.1 Tools & Libraries Used

Tool / Library	Version	Purpose
Python	3.11+	Primary analysis language
Pandas	2.x	Data ingestion, cleaning, aggregation
scikit-learn	1.x	StandardScaler, KMeans, PCA, metrics
SciPy	1.x	Hierarchical clustering, dendrogram
Matplotlib + Seaborn	3.x	EDA and static visualisations
Plotly	5.x	Interactive visualisations in Streamlit
Streamlit	1.x	Interactive web dashboard

9.2 Feature Engineering Summary

Feature	Source	Encoding	Notes
age	date_of_birth	Derived (numeric)	Days from DOB to 2024-01-01
satisfaction_score	clients.csv	Numeric (1–5)	Used as-is
num_properties	properties.csv	Aggregated count	Sold listings per client
total_investment	properties.csv	Aggregated sum	Sum of sale_price per client
avg_unit_area	properties.csv	Aggregated mean	Mean floor_area_sqft per client
loan_enc	loan_applied	Label Encoded (0/1)	Yes=1, No=0
type_enc	client_type	Label Encoded (0/1)	Company=1, Individual=0
purpose_enc	acquisition_purpose	Label Encoded (0/1)	Investment=1, Home=0
country_*	country	One-Hot Encoded	10 binary columns
referral_*	referral_channel	One-Hot Encoded	3 binary columns

9.3 Cluster Centroid Profiles

Metric	Luxury	Global	Corporate	First-Time
Count	7	74	458	316
Avg Age	48.6	49.8	52.8	53.6
Avg Satisfaction	2.57	2.88	3.02	3.09
Avg Investment	\$1,568,590	\$1,272,131	\$1,265,070	\$1,260,165
Avg Unit Area (sqft)	1,369	1,166	1,151	1,151
Loan Rate	29%	38%	38%	36%
Investment Purpose %	14%	34%	31%	32%